

Effects of in-situ admixtures and simulated Martian regolith on the mechanical properties and microstructure of basic magnesium sulfate cement for potential martian construction

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ABSTRACT

With the continuous advancement of Mars exploration, in-situ resource utilization (ISRU) for Martian infrastructure has become a research hotspot. Basic magnesium sulfate cement (BMSC) has the potential to be fully prepared from Martian resources, which suggests that it could be a promising material for Martian construction. In this study, a conceptual ISRU pathway based on Martian magnesium sulfate minerals and phosphate-bearing regolith is proposed, and its material feasibility was preliminarily evaluated under terrestrial laboratory conditions using a self-prepared simulated Martian regolith system. Reactive MgO, MgSO₄, and simulated regolith powder were used to prepare a BMSC-Martian regolith composite system (BMSC-MS). Two potentially in-situ obtainable additives, phosphoric acid and citric acid-based beverage powder, were introduced to investigate their effects on mechanical properties, hydration products, microstructure evolution, thermal stability, and pore structure. The results show that phosphoric acid exhibits significantly better regulation performance than the citric acid-based additive. At an optimal dosage of 1 wt%, the BMSC-MS system achieved the best overall performance, with a 28-day compressive strength of 39.09 MPa. Microstructural analyses indicate that phosphoric acid promotes the formation of the 5-1-7 phase and amorphous M-(A-)S-H gels, suppresses disordered Mg(OH)₂ precipitation, and improves matrix densification through pore refinement. Mercury intrusion porosimetry further confirmed a reduced total porosity of 20.89% in the optimized system. These findings provide a preliminary material-level assessment of the feasibility of BMSC-based composites derived from a conceptual Martian ISRU route and offer preliminary theoretical support for future extraterrestrial construction material design.

1. Introduction

From Kepler's first observation of the Martian orbit to the earliest images of Martian landscapes and the successful landing of rovers, Mars has always embodied humanity's aspiration for interplanetary exploration. As the planet most similar to Earth in the solar system, Mars features a thin atmosphere, clear evidence of water resources, stable geology, and a diurnal cycle close to that of Earth, making it one of the most promising extraterrestrial bodies for habitability. In recent years, the United States, China, and the European Space Agency have successively launched manned Mars mission plans, marking a transition of Martian exploration from unmanned detection to crewed landing. The

establishment of Martian bases will provide fundamental platforms for scientific exploration, life support, resource utilization, and deep-space transportation, serving as a cornerstone for long-term human settlement and the creation of extraterrestrial civilization.

Cementitious materials are increasingly recognized as indispensable for the infrastructural development of lunar and Martian outposts, providing essential support for sustained extraterrestrial exploration [1]. Recent comprehensive reviews corroborate this paradigm, establishing space-grade concrete systems as the preeminent candidates for extraterrestrial habitats owing to their structural versatility and radiation shielding capabilities [2]. To mitigate the severe operational constraints of extraterrestrial environments, contemporary vanguard

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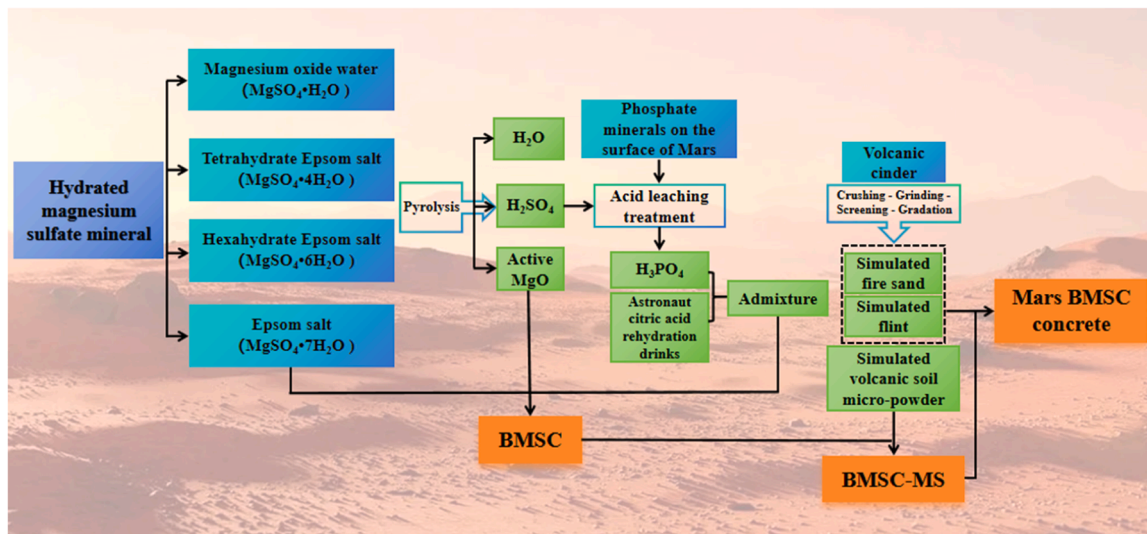


Fig. 1. Preparation pathway of Martian BMSC concrete based on ISRU.

research has prioritized the development of highly specialized material formulations. Notably, the synthesis of geopolymer-based high-strength lunar concrete [3] and the systematic evaluation of its tensile performance under cryogenic conditions [4] have robustly demonstrated the structural viability of these systems. Furthermore, to decouple construction logistics from terrestrial supply chains, recent advancements in waterless concrete architectures [5] and cement-free Martian matrices engineered for severe diurnal thermal cycling in CO₂-rich atmospheres [6] represent the current state-of-the-art. Given the prohibitive transportation costs of delivering materials from Earth, utilizing Mars' abundant natural resources through In-Situ Resource Utilization (ISRU) is the optimal and indispensable solution.

Nevertheless, despite these promising technological trajectories, implementing existing ISRU-based material systems on Mars remains impeded by critical structural and chemical bottlenecks. Specifically, Vitse et al. [7] developed a geopolymer-based Martian concrete using Martian regolith simulant and metakaolin. However, this system heavily depends on external alkali activators (e.g., NaOH, LiOH) that are difficult to obtain from Martian resources and require energy-intensive extraction. Troemner et al. [8] proposed a sulfur-regolith composite (Marscrete). Recent thermomechanical assessments and numerical simulations of sulfur-based extraterrestrial materials [9] reveal critical vulnerabilities: sulfur must be melted at elevated temperatures (140–160 °C), exhibits poor long-term stability under Mars' extreme diurnal temperature variations (−100 °C to 20 °C), and lacks readily accessible high-purity deposits. Qiao et al. [10] explored a cement-free approach by compacting regolith simulant directly into bricks, though such physically compacted materials lack effective cementitious phases. Gupta et al. [11] fabricated “Space Bricks” via high-temperature sintering at 1100–1150 °C; while simplifying raw materials, these ceramic-like products are vulnerable to microcrack propagation under Martian low-pressure environments. Furthermore, innovative bio-polymer binders like AstroCrete [12] raise ethical concerns, critically depend on limited physiological resources, and lack industrial scalability. Consequently, a critical research gap persists in developing a high-performance binder that maximizes the utilization of locally available Martian resources while ensuring environmental resilience without relying on extreme thermal processing.

Sulfate minerals are common evaporite deposits on the Martian surface, characteristically incorporating various forms of crystalline water within their structures. Recent high-resolution orbital spectroscopy combined with in-situ rover explorations has conclusively identified abundant Ca-rich sulfates (e.g., gypsum) and diverse hydrated

magnesium sulfates across widespread Martian terrains, such as Gale and Jezero craters [13,14]. Furthermore, the latest in-situ analyses from the Chemistry and Mineralogy (CheMin) instrument onboard the Curiosity rover and the Planetary Instrument for X-ray Lithochemistry (PIXL) onboard the Perseverance rover have revealed that the Martian regolith is primarily composed of primary igneous minerals (including olivine, pyroxene, plagioclase, Fe-Ti-Cr spinels, and phosphates) and diverse secondary alteration minerals (such as amorphous silica, hematite, nanophase oxides, clays, and sulfates) [15,16].

The abundant calcium sulfates (e.g., gypsum) and magnesium sulfates present on the Martian surface can be converted into reactive cementitious precursors through relatively simple, low-energy thermochemical processes (i.e., low-temperature calcination), presenting a highly viable technological pathway for developing Martian concrete. Within the sulfate-based binder family, both gypsum cement and basic magnesium sulfate cement (BMSC) represent mature, low-temperature-calcined inorganic binders that have played significant roles in terrestrial construction. However, traditional gypsum cement exhibits intrinsically low mechanical strength, restricting its use primarily to non-structural or decorative applications. In striking contrast, BMSC has recently emerged as a highly promising, low-carbon binder capable of effectively binding aggregates to produce robust, load-bearing structural concrete [17]. Recent microstructural investigations have elucidated the highly stable phase composition of the MgO-MgSO₄-H₂O system [18], suggesting that Martian magnesium sulfate minerals may provide a promising material pathway for future ISRU-oriented extraterrestrial construction.

In ISRU pathways, although admixtures are used in minimal quantities, they play an irreplaceable role in regulating early hydration. Recent advancements demonstrate that organic molecules and specific inorganic acids can fundamentally promote the formation of the robust 5-1.7 phase in BMSC, significantly improving microstructural compactness and early strength [18,19]. However, given that the extreme Martian conditions, including low temperature, low pressure, and high radiation, severely hinder normal hydration processes, such active regulation via specific admixtures becomes exceptionally crucial. Therefore, this study proposes a dual-pathway strategy for admixtures: (i) producing inorganic phosphoric acid admixtures directly from Martian phosphate minerals, and (ii) utilizing organic acids (e.g., citric acid) recycled from astronauts' rehydration beverages, achieving both performance regulation and resource recycling.

Based on this concept, the present study systematically investigates the effects of these two in-situ admixtures on the reaction mechanisms,

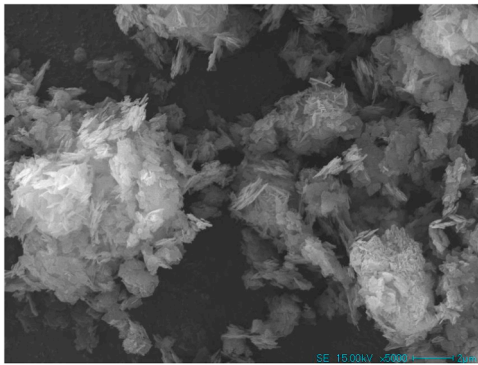


Fig. 2. SEM image of T600.

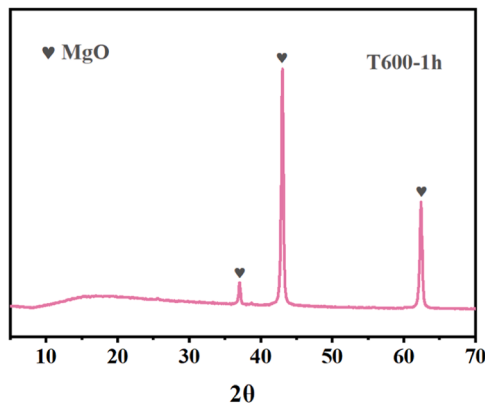


Fig. 3. XRD analysis of T600.

2. Conceptual ISRU pathway for Martian BMSC and preliminary energy considerations

To explore the potential of ISRU for Martian infrastructure, this study proposes a conceptual preparation route for BMSC and its concrete using extractable Martian surface minerals, as illustrated in Fig. 1.

Initially, utilizing the widely distributed hydrated magnesium sulfate minerals on the Martian surface as fundamental raw materials, reactive magnesium oxide (MgO), water, and sulfuric acid are obtained through thermal decomposition. Among these products, reactive MgO serves as the core precursor for preparing BMSC; the water released during pyrolysis can be utilized as concrete mixing water, providing an essential supply for the water-deficient Martian environment; and the generated sulfuric acid can be deployed for admixture preparation, mineral processing, and further resource utilization. Simultaneously, the Martian surface regolith is rich in forsterite (Mg_2SiO_4), which can be leached by sulfuric acid to obtain magnesium hydroxide [20], and subsequently converted into highly reactive MgO via low-temperature calcination, thereby providing supplementary raw materials for the BMSC system.

Based on the confirmed detection of phosphate minerals on the Martian surface [15,16,21–24], this study proposes a "dual-pathway" admixture strategy. On the one hand, the system-generated sulfuric acid is utilized for the wet acid leaching of Martian phosphate minerals to extract inorganic phosphoric acid as a regulating admixture. On the other hand, rehydrated beverages (rich in organic components such as citric acid) from astronaut systems are recovered to serve as organic admixtures.

Among current construction material strategies based on Martian in-situ resources, sintering or melting-based routes generally require heating bulk regolith to 1100–1500 °C, where the high-temperature process involves the majority of the material mass, significantly increasing overall system energy consumption and thermal management burdens [25]. In contrast, in the BMSC-MS system, regolith powder can directly function as the principal aggregate and mineral component participating in the cementitious reaction, without the need for high-temperature activation or complex chemical modification. Previous studies have shown that, under optimized mix proportions, the regolith content can reach 84.75%, while the 28-day compressive strength of BMSC-regolith concrete remains as high as 59.2 MPa [26], indicating the potential for a high level of ISRU. In this system, high-temperature processing is limited to the preparation of a relatively small fraction of MgO precursor, whereas regolith utilization and cementitious reactions proceed at ambient temperature. From a system-level perspective, this process route therefore holds potential for reducing overall energy demand.

Exploratory experimental results indicate that the magnesium oxide produced by one-step calcination at 1150 °C for 3 h exhibits low reactivity. Consequently, this study adopts a 'washing-hydration-600 °C light-calcination' two-step process to successfully prepare highly reactive MgO (denoted as T600), whose micro-characteristics are shown in Fig. 2 and Fig. 3. Since the laboratory-scale preparation yield of T600 remains limited, the subsequent systematic experiments were conducted using commercially available light-burned MgO. As demonstrated by the comparative physicochemical characterization presented in Section 2.1 (Table 5), the commercial MgO exhibits comparable reactivity and key physicochemical properties to T600, thereby ensuring the consistency and reliability of the subsequent experimental investigation under terrestrial laboratory conditions.

3. Materials and experimental methods

3.1. Experimental preparation

3.1.1. Raw materials

Simulated Martian Regolith. The simulated Martian regolith was sourced from volcanic scoria collected at Dayishan, Huinan County, Jilin

Table 1
Physical properties and chemical composition of volcanic cinder.

NO.	Item	Test standard	Result
1	Bulk density (kg/m^3)	GB/T17431.2–2010	640
2	Apparent density (kg/m^3)	GB/T17431.2–2010	1322
3	Porosity (%)	GB/T17431.2–2010	52
4	Cylinder compressive strength (MPa)	GB/T17431.2–2010	4.2
5	SiO_2 (%)	JC/T874–2009	49.54
6	Al_2O_3 (%)	JC/T874–2009	15.46
7	Fe_2O_3 (%)	JC/T874–2009	12.77
8	MgO (%)	JC/T874–2009	7.74
9	CaO (%)	JC/T874–2009	8.2
10	K_2O (%)	JC/T874–2009	2.4
11	Na_2O (%)	JC/T874–2009	3.14
12	Radioactivity	Internal exposure index (IRa)	0.16
		External exposure index (Ir)	0.68

mechanical properties, and microstructural evolution of the BMSC-Martian regolith micro-powder (BMSC-MS) system. This work aims to bridge the aforementioned research gap by first evaluating the material-level feasibility of a conceptual ISRU-oriented BMSC-MS system under terrestrial laboratory conditions, thereby providing a theoretical and technical basis for subsequent studies on advanced extraterrestrial construction materials.



Fig. 4. Field photograph of volcanic scoria.



Fig. 5. Physical appearance of NUAA-1M simulated Martian regolith powder.

Province. Its mineralogical composition and macroscopic morphology are presented in Table 1 and Fig. 4, respectively. Following crushing and grinding processes, fractions with different particle sizes were obtained, including regolith micropowder, Martian sand, and Martian gravel, which were used as mineral admixture and fine/coarse aggregates for

Martian cement and concrete, respectively. This integrated material system was designated as “NUAA-1M”, which was developed to simulate the basaltic regolith extensively distributed across the Martian surface. In this study, the fraction with particle size below 70 μm was selected as the mineral admixture to simulate the reactive portion of regolith capable of participating in cementitious reactions, thereby establishing the Martian BMSC-MS system. The physical appearance of the NUAA-1M regolith micropowder is shown in Fig. 5. Zhang et al. [27] compared the composition of NUAA-1M with that of Martian soil and various widely used Martian simulants, demonstrating a high degree of chemical similarity to authentic Martian regolith and compositional equivalence to other representative simulants.

The particle size distribution and cumulative distribution of NUAA-1M micropowder were characterized using laser diffraction analysis (Fig. 6). The average particle size was approximately 60.46 μm , with a median diameter of 42.43 μm . The chemical composition of NUAA-1M simulant powder was analyzed by X-ray fluorescence (XRF) and benchmarked against the actual Martian regolith data recorded by the Curiosity rover at Gale Crater (Table 2). The results demonstrate a high degree of compositional similarity between the two. The combined content of SiO_2 , Al_2O_3 , and Fe_2O_3 reaches 73.41%, which exceeds the

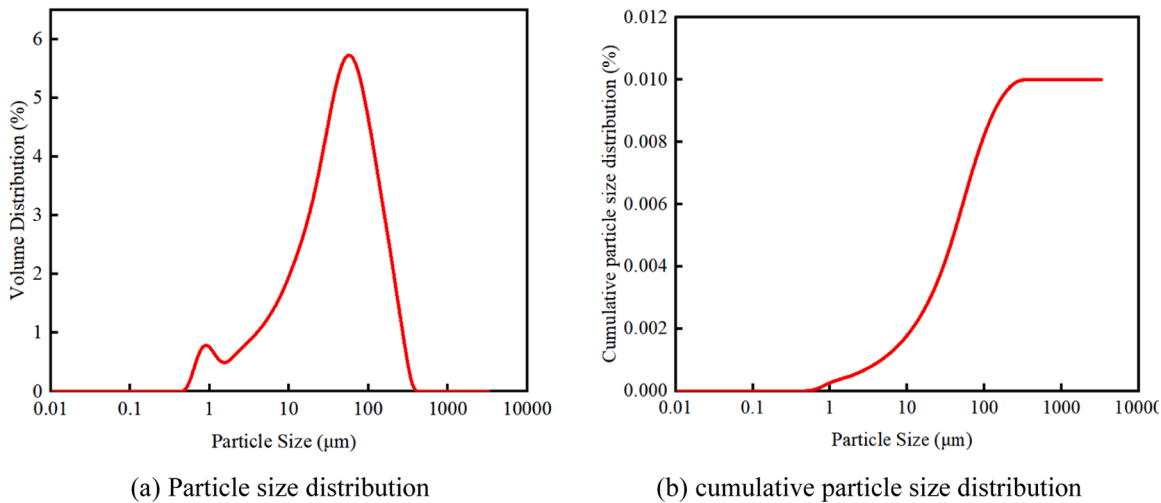


Fig. 6. Particle size distribution and cumulative particle size distribution curves of NUAA-1M simulated Martian regolith powder.

Table 2
Comparison of chemical composition between NUAA-1M Martian regolith simulant powder and actual Martian regolith (Curiosity) (wt%).

Component	Al_2O_3	CaO	Fe_2O_3 / FeO	MgO	K_2O	SiO_2	Na_2O	Others**
NUAA-1M	14.72	6.17	8.96	4.29	2.40	49.73	4.68	9.04
Actual Martian Regolith (Curiosity) [27]	9.43	7.28	19.19*	8.69	0.49	42.88	2.72	9.32

* Note: Data for Curiosity in Ref. [27] are reported as FeO.
** Note: Including TiO_2 (1.19%), MnO (0.41%), P_2O_5 (0.94%), SO_3 (5.45%), Cl (0.69%), etc.

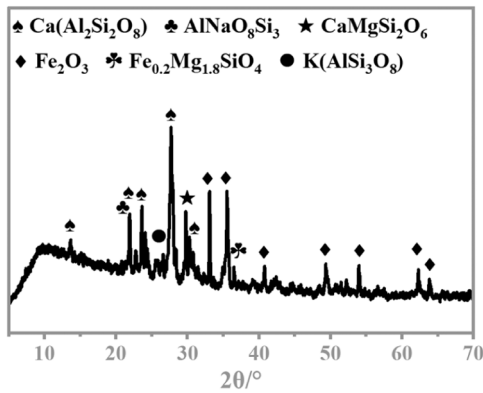


Fig. 7. XRD pattern of NUAA-1M simulated Martian regolith powder.

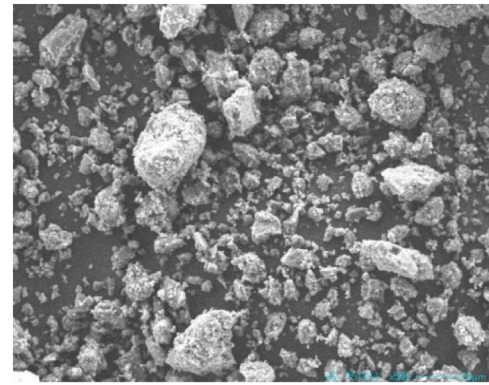


Fig. 8. Microstructure of NUAA-1M simulated Martian regolith powder.

minimum requirement (70%) specified for Class N fly ash in ASTM C618. The X-ray diffraction (XRD) pattern of NUAA-1M micropowder (Fig. 7) indicates that the crystalline phases are primarily composed of feldspar, hematite, olivine, and diopside. Quantitative phase analysis is provided in Table 3, where 20 wt% α - Al_2O_3 was introduced as an internal standard for the determination of amorphous content. Scanning electron microscopy (SEM) images (Fig. 8) reveal irregular particle morphology with rough surfaces. The Brunauer-Emmett-Teller (BET) specific surface area is $2.03 \text{ m}^2/\text{g}$, indicating a certain potential for cementitious reactivity.

Furthermore, Zhang et al. [27] systematically investigated the pozzolanic reaction behavior of NUAA-1M in the basic magnesium sulfate cement system. Through quantitative XRD, SEM-EDS NMR, and XPS analyses, they confirmed that the amorphous phases and partially reactive aluminosilicate components in the simulant can participate in alkaline reactions, leading to the formation of amorphous hydration products such as M-(A)-S-H gel, thereby demonstrating distinct pozzolanic activity. Therefore, NUAA-1M not only satisfies the compositional criteria for pozzolanic materials, but its reactivity has also been experimentally validated. On this basis, the present study further investigates the regulatory mechanisms of different in-situ additives on the activation pathways of regolith reactivity and the associated microstructural evolution.

Light-burned magnesium oxide. The light-burned magnesium oxide (MgO) powder was prepared by calcining magnesite at $850\text{--}900^\circ\text{C}$ followed by grinding to achieve a fineness of 200 mesh. Its chemical composition is listed in Table 4. According to the water absorption method [28], the content of active MgO in the powder was 80 wt%, and its bulk density was 0.65 g/mL .

To ensure the accuracy and rigor of the experimental design, the key properties of the light-burned MgO used in this study were compared with those of T600 (Table 5). The results indicate that both samples exhibit comparable citrate colorimetric time, reactive MgO content (determined by the hydration method), and loss on ignition. Although differences in grain size are observed, their overall physicochemical characteristics are similar. These results support the suitability of the light-burned MgO as a reactive precursor in the present study.

Chemical admixtures and other materials. In addition to the primary cementitious materials and the simulated Martian regolith, several auxiliary reagents were used in this study. Food-grade magnesium sulfate heptahydrate ($\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$, purity $\geq 99.0\%$, Jiangsu Kolod Food Ingredients Co., Ltd., China) was employed as the sulfate component. The admixtures included analytical-grade phosphoric acid (H_3PO_4 ,

purity $\geq 85\%$, Guangdong Xilong Scientific Co., Ltd., China) and a commercial lemon powder (True Citrus, USA). The lemon powder mainly consisted of citric acid, lemon oil, and concentrated lemon juice, and was specifically selected to simulate the rehydrated beverage system for astronauts. In addition, analytical-grade citric acid monohydrate ($\text{C}_6\text{H}_8\text{O}_7 \cdot \text{H}_2\text{O}$, purity $\geq 99.5\%$, Xi'an Tianmao Baoding Biotechnology Co., Ltd., China), which is a traditionally used conventional admixture in the BMSC system, was employed as a control group for comparison with the two admixture systems proposed in this study. Furthermore, 200-mesh corundum (α - Al_2O_3) was used as the internal standard for the quantitative determination of amorphous phase content, and all mixing procedures were carried out using drinking water that complies with the national standard.

3.1.2. Mix proportion

Previous studies have shown that the regolith content in lunar and Martian construction materials is generally controlled within 60–70% [29,30]. Based on this, the dosage of NUAA-1M simulated Martian regolith powder was fixed at 60 wt% of the total cementitious materials to ensure comparability and rationality of the results. Combined with preliminary trial experiments, it was found that under this dosage, when the molar ratio of MgO to $\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$ was 3, the slurry could not maintain adequate formability for subsequent mechanical testing. Therefore, to ensure the formability of the slurry, the minimum molar ratio was set at 4, and 5 and 6 were further selected as comparative groups to systematically investigate the effects of two types of admixtures, inorganic phosphoric acid and citric acid powder, on the mechanical properties and microstructure of the BMSC-MS system. Considering that this study aims to maximize the utilization of hydrated magnesium sulfate resources potentially available on Mars, higher molar ratios such as 7 were not further adopted once satisfactory formability and mechanical performance were achieved.

In addition, the selected admixture system was compared with the widely used conventional admixture citric acid in BMSC to evaluate differences in performance and regulation effects. Five curing ages were set, and three parallel tests were conducted for each mix proportion, as detailed in Table 6.

3.1.3. Preparation method

According to the designed mix proportions, the pre-weighed $\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$ and the selected additives (H_3PO_4 , lemon powder, or citric acid) were first dissolved in water. The solution was stirred with a glass rod to form a homogeneous magnesium sulfate solution and then

Table 3
Quantitative phase composition of NUAA-1M simulated Martian regolith powder.

Component	$\text{AlNaO}_8\text{Si}_3$	$\text{Ca}(\text{Al}_2\text{Si}_2\text{O}_8)$	$\text{CaMgSi}_2\text{O}_6$	Fe_2O_3	$\text{Fe}_{0.2}\text{Mg}_{1.8}\text{SiO}_4$	$\text{K}(\text{AlSi}_3\text{O}_8)$	Amorphous phase
Mass fraction/%	24	26.7	13.4	6.7	3.5	14.5	11.2

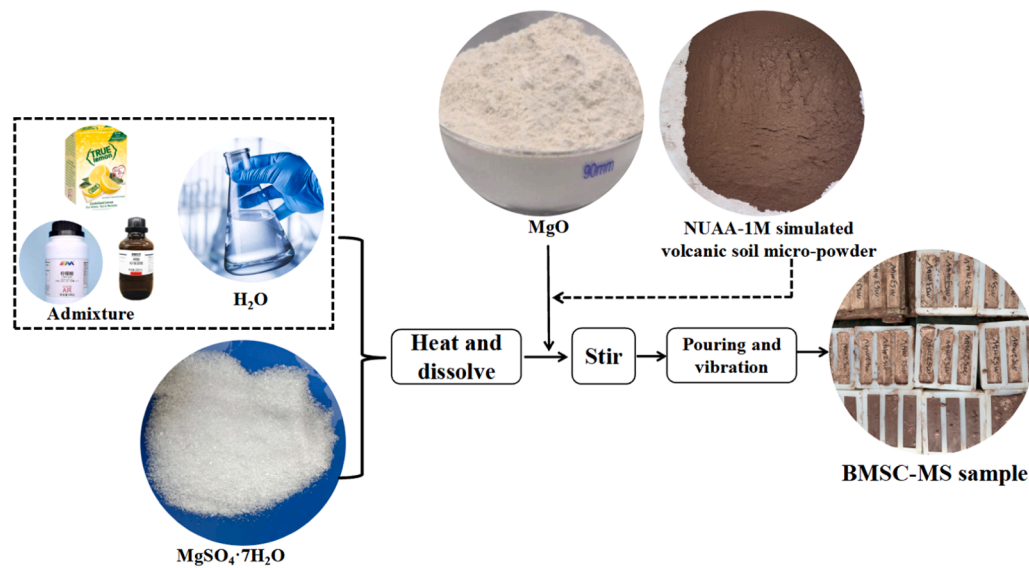


Fig. 9. Preparation process of BMSC-MS specimens.

Table 4
Chemical composition of magnesium oxide.

Component	MgO	Fe ₂ O ₃	CaO	MgCl ₂	MgSO ₄	TiO ₂	MnO	P ₂ O ₅	LOI
Wt /%	95.72	1.21	1.20	0.25	0.26	0.48	0.79	0.09	1.71

Note: LOI refers to loss on ignition.

poured into a mixing bowl. The mixing bowl was placed on a cement paste mixer and stirred at low speed, while the lightly calcined MgO powder and NUAA-1M simulated Martian regolith fine powder were added sequentially within 30 s. The mixer was then switched to high-speed stirring for 30 s. After scraping off the residual paste on the blades and mixing bowl walls, high-speed stirring continued for 1 min to obtain a uniform cement paste.

The prepared slurry was immediately poured into molds with dimensions of 20 mm × 20 mm × 80 mm, compacted on a vibration table for 60 s, leveled, and then sealed with plastic film. Considering that the present system does not contain conventional sand or coarse aggregates and should therefore be classified as a paste-based cementitious system with simulated Martian regolith micropowder as the functional mineral component, and that the BMSC-MS system exhibits rapid early-age hydration reactions, a specimen size of 20 mm × 20 mm × 80 mm was adopted in this study with reference to previous related studies [31,32], so as to reduce the potential risk of internal temperature gradients and shrinkage-induced cracking during early curing.

In addition, since the maximum particle size of all solid constituents in the system is within the micrometer scale, the selected specimen dimensions are considered sufficiently larger than the characteristic particle scale for comparative macroscopic mechanical testing [33,34]. Consequently, no significant size effect is expected to be introduced into

Table 5
Physical and chemical properties of MgO samples.

Sample	Loss on ignition /%	Grain size /nm	Citrate colorimetric time /s	Reactive MgO content /%
T600	25.4	13.6	29.81	73.08
Light-burned MgO (raw material)	25.03	44.87	31.62	79.85

the comparative strength results among different mix proportions.

After casting, the specimens were cured in a standard humidity chamber for 24 h before demolding, followed by continued curing at room temperature until the designated testing ages.

3.2. Analytical methods

3.2.1. Basic performance tests of BMSC-MS

The setting time of BMSC-MS pastes was determined using a Vicat apparatus in accordance with GB/T 1346–2011 [35]. The initial and final setting times were recorded, with the moment of water addition defined as time zero.

The fluidity of the fresh BMSC-MS pastes was measured following GB/T 2419–2005 [36]. A standard truncated cone mold (top diameter 36 mm, bottom diameter 60 mm, height 60 mm) was employed. The spread diameter of each mixture was recorded after 25 drops on a flow table.

3.2.2. Mechanical performance testing

The compressive and flexural strengths of BMSC-MS specimens were measured using HG-YH100D and YAW-300D testing machines. The HG-YH100D testing machine has a load capacity of 100 kN/5 kN and a sensitivity of 0.01 kN/0.01 kN. The loading rates for compressive and flexural strength tests were 2.4 kN/s and 50 N/s, respectively.

Following GB/T 17671–2021 [37], the flexural and compressive strengths of BMSC-MS were determined. The flexural strength of BMSC-MS is calculated using the following formula:

$$f_f = \frac{1.5 F_f L}{b^3} \quad (1)$$

Where: f_f — flexural strength of BMSC-MS (MPa);

F_f — load at specimen fracture (N);

L — span between the two support points (mm), set as 50 mm;

b — side length of the square cross-section of the specimen (mm), set

Table 6

Mix Proportion Design of BMSC-MS with Different Additive Types and Dosages.

NO.	^a M/S Molar Ratio	Mass of Each Component in the Paste / g						w/b ^b
		MgO	MgSO ₄ ·7 H ₂ O	NAAA-1M Simulated Martian Regolith	H ₃ PO ₄	Lemon powder	CaH ₈ O ₇ ·H ₂ O	
P0.2M4	4	246	377	646	0.49	0	0	0.33
C0.2M4					0	0	0.49	
L0M4					0	0	0	
L0.1 M4					0	0.25	0	
L0.2M4					0	0.49	0	
L0.3M4					0	0.74	0	
L1M4					0	2.5	0	
L2M4					0	4.92	0	
P1M4					2.5	0	0	
P2M4					4.92	0	0	
P0.2M5	5	269	330	646	0.54	0	0	0.33
C0.2M5					0	0	0.54	
L0M5					0	0	0	
L0.1M5					0	0.27	0	
L0.2M5					0	0.54	0	
L0.3M5	6	287	293	646	0	0.81	0	0.33
P0.2M6					0.57	0	0	
C0.2M6					0	0	0.57	
L0M6					0	0	0	
L0.1M6					0	0.29	0	
L0.2M6					0	0.57	0	
L0.3M6					0	0.86	0	

Note: ^a M/S, i.e., the molar ratio of MgO to MgSO₄·7H₂O; ^b w/b, i.e., water-to-binder ratio; the same below.

as 20 mm.

The compressive strength is calculated using the following formula:

$$f_{cu} = \frac{F_c}{A} \quad (2)$$

Where: f_{cu} — compressive strength of BMSC-MS (MPa);

F_c — load at specimen failure (N);

A — compressed area of the specimen (mm²), set as 400 mm².

The splitting tensile strength is calculated using the following formula:

$$f_{st} = \frac{0.637F_s}{A} \quad (3)$$

Where: f_{st} — splitting tensile strength of BMSC-MS (MPa);

F_s — load at specimen fracture (N);

A — compressed area of the specimen (mm²), set as 400 mm².

3.2.3. Phase composition and microstructure analysis

Sample Collection. The powdered and bulk specimens obtained

after crushing were immediately immersed in isopropanol to terminate hydration. After one day, the isopropanol was replaced with fresh isopropanol and the samples were soaked for a total of seven days. The powder samples after seven days of immersion were dried in a vacuum oven at 40 °C for 48 h.

X-ray Diffraction (XRD) Testing. The dried samples were mixed with α -Al₂O₃ at a mass ratio of 4:1, ground into fine powder, and passed through a 75 μ m sieve (200 mesh, No. 200) to obtain the powdered specimen. XRD measurements were conducted using a D/max-2500PC diffractometer (PANalytical, Netherlands) with Cu/K α radiation (λ = 0.15419 nm), an accelerating voltage of 15 kV, a 2 θ scanning range of 5–70°, and a scanning rate of 2°/min. Based on the XRD data, quantitative mineral analysis of BMSC-MS was performed using Topas 5.0 software (Bruker, Germany) according to the Topas user manual [38].

Microstructure Analysis. For electron microscopy, bulk samples of approximately 5 mm were gold-coated and analyzed using an EPMA-8050G field-emission electron probe micro-analyzer for micro-region composition and morphology. The gold sputtering conditions were: current of 10 mA, pressure of 0.2 mbar (20 Pa), and sputtering thickness of 20 nm. Micro-region elemental analysis was further performed using an INCAEnergy Energy Dispersive Spectrometer (EDS, Oxford Instruments, UK).

Thermal Analysis. Thermogravimetric-differential thermal analysis (TG-DTA) was used to investigate the thermal decomposition of BMSC-MS. Powdered samples were placed in an HCT-1 microcomputer-controlled differential thermal analyzer, and the analysis was performed under a nitrogen atmosphere from 25 to 1050 °C with a heating rate of 10 °C/min.

Pore Structure Analysis. Before pore characterization, the samples were vacuum-dried and broken into fragments of approximately 5 mm. Mercury intrusion porosimetry was conducted using an AutoPore IV 9500 high-performance fully automatic mercury porosimeter (Micro-meritics, Shanghai) to measure the pore structure characteristics. The tested pore diameter range was 0.003–1000 μ m, with a maximum applied pressure of 208 MPa and a contact angle of 130°.

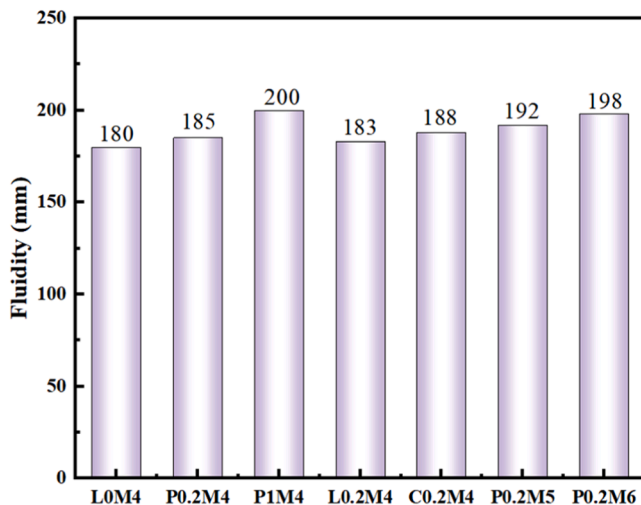


Fig. 10. Fluidity of BMSC-MS pastes.

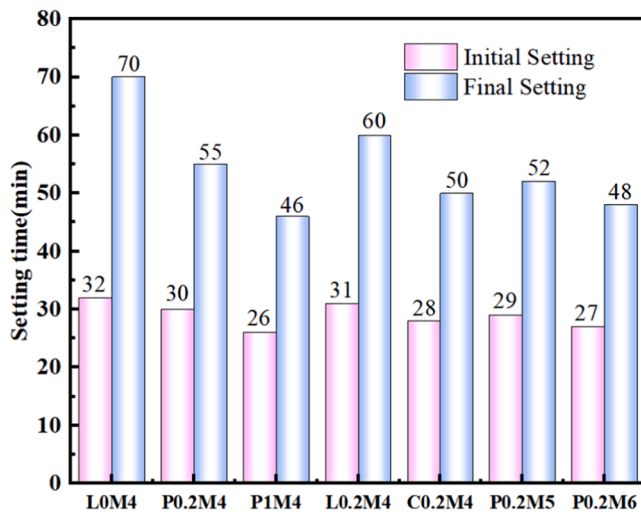


Fig. 11. Setting time of BMSC-MS pastes.

4. Results and discussion

4.1. Fluidity and setting time of BMSC-MS pastes

Fluidity is a key parameter for evaluating the workability of cementitious pastes, while setting time directly determines the available operational window. Investigation of the rheological and setting behavior of BMSC-MS pastes not only provides a theoretical basis for performance regulation, but also supports their potential application in in-situ construction scenarios on Mars, such as additive manufacturing.

Fig. 10 presents the fluidity results of BMSC-MS pastes with different mix proportions. All mixtures exhibited spread diameters in the range of 180–210 mm, indicating satisfactory workability. As the M/S molar ratio increased from 4 to 6, a slight increase in fluidity was observed. For instance, under the condition of 0.2% phosphoric acid addition, the spread diameter increased from approximately 185 mm to 198 mm. This suggests that a higher MgO proportion may improve particle dispersion, thereby enhancing flowability.

Both the type and dosage of additives exerted a noticeable influence on fluidity. In the reference mixture with M/S = 4 (L0M4), the spread diameter was approximately 180 mm. The incorporation of 0.2% phosphoric acid or citric powder resulted in improved flowability, while increasing the phosphoric acid dosage to 1% further enhanced the spread to approximately 200 mm. This behavior may be attributed to the effective retardation of rapid MgO hydration in the early stage, particularly in the presence of phosphoric acid, which suppresses the premature formation of $\text{Mg}(\text{OH})_2$ gel and maintains paste plasticity.

Such regulation provides favorable conditions for the uniform nucleation and growth of the 5-1-7 phase at later stages.

Fig. 11 shows the initial and final setting times measured using a Vicat apparatus. The initial setting times of all mixtures were primarily distributed between 20 and 32 min, while final setting times ranged from 35 to 70 min, indicating an appropriate operational time window. The introduction of additives enabled effective regulation of setting behavior. With 0.2% phosphoric acid addition, the initial and final setting times were approximately 30 min and 55 min, respectively. When the dosage was increased to 1%, the initial setting time decreased to 26 min and the final setting time to approximately 46 min, suggesting a slight accelerating effect. A similar trend was observed for the citric powder system, although the magnitude of regulation was less pronounced than that of phosphoric acid. The moderate shortening of setting time is consistent with the mechanism whereby additives facilitate early ionic migration and accelerate the nucleation of hydration products.

Compared to that of traditional cast-in-place concrete on Earth, the setting time of the present system is relatively short. However, this rapid setting characteristic may be potentially advantageous for future Martian in-situ construction scenarios, particularly for additive manufacturing applications. On the one hand, future Martian habitat construction relies heavily on automated rapid prototyping technologies such as 3D printing. A short initial setting time endows the material with excellent early-stage buildability, ensuring that the extruded bottom layers rapidly solidify and gain sufficient yield stress to support the weight of continuous upper layers without collapsing. On the other hand, the Martian surface is characterized by extremely low atmospheric pressure, where liquid water is highly susceptible to rapid boiling or sublimation. This rapid setting behavior may help mitigate potential risks associated with water loss under low-pressure extraterrestrial environments, thereby preventing structural failure due to dehydration.

4.2. Effect of different types and dosages of admixtures on the mechanical properties of BMSC-MS

Fig. 12 presents the evolution of compressive strength, flexural strength, and splitting tensile strength of BMSC-MS samples at different ages under an M/S ratio of 4. Overall, all three mechanical properties increased significantly with age, indicating ongoing hydration reactions and progressive densification of the internal structure. It can be observed that samples containing admixtures exhibited higher mechanical strength than the control group at all ages, demonstrating that the admixtures promoted the formation of hydration products and enhanced the mechanical performance of the BMSC-MS system.

At the same dosage (0.2%), the C0.2M4 sample showed a 28-day compressive strength of 26.49 MPa and relatively high flexural and

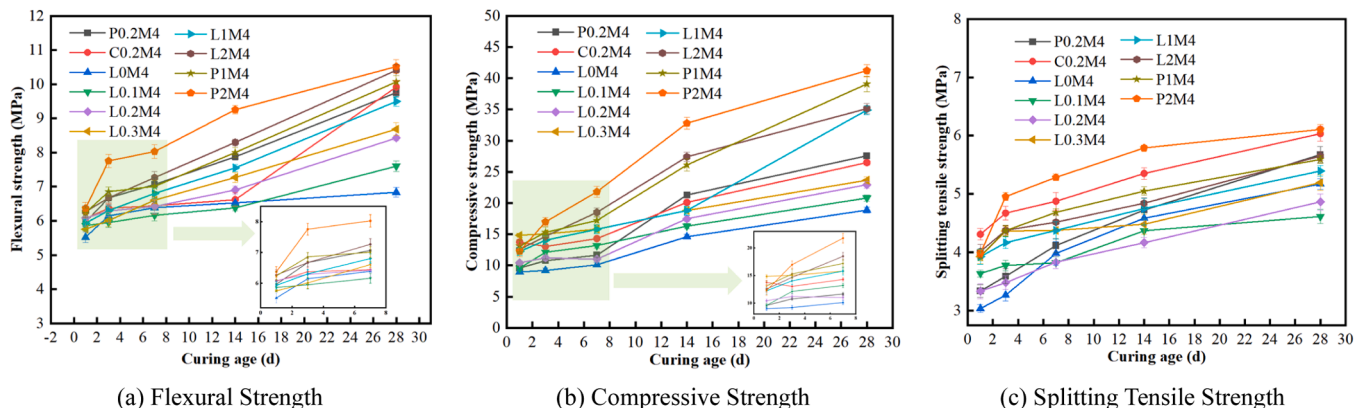


Fig. 12. Effect of Admixture Type and Dosage on the Mechanical Properties of BMSC-MS (M/S = 4).

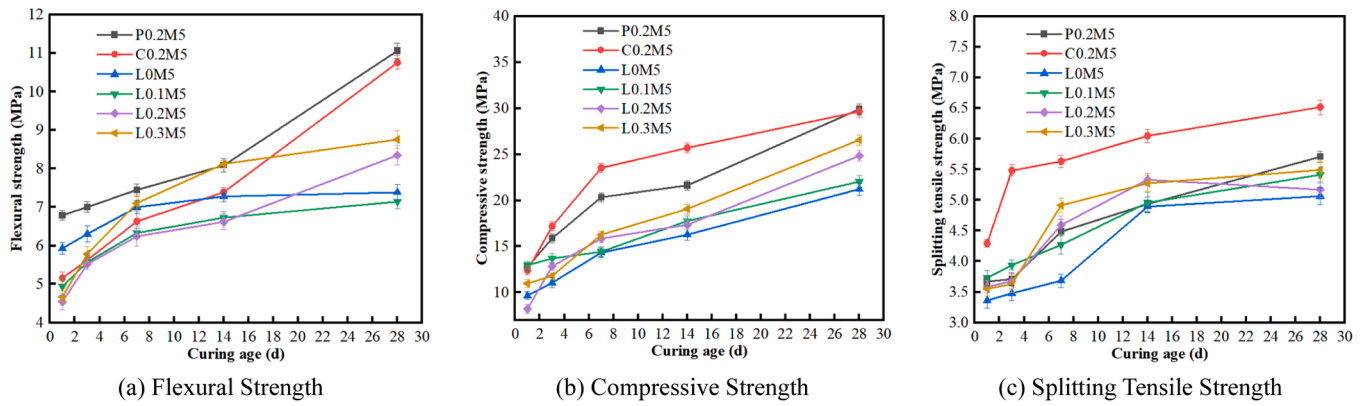


Fig. 13. Effect of Admixture Type and Dosage on the Mechanical Properties of BMSC-MS (M/S = 5).

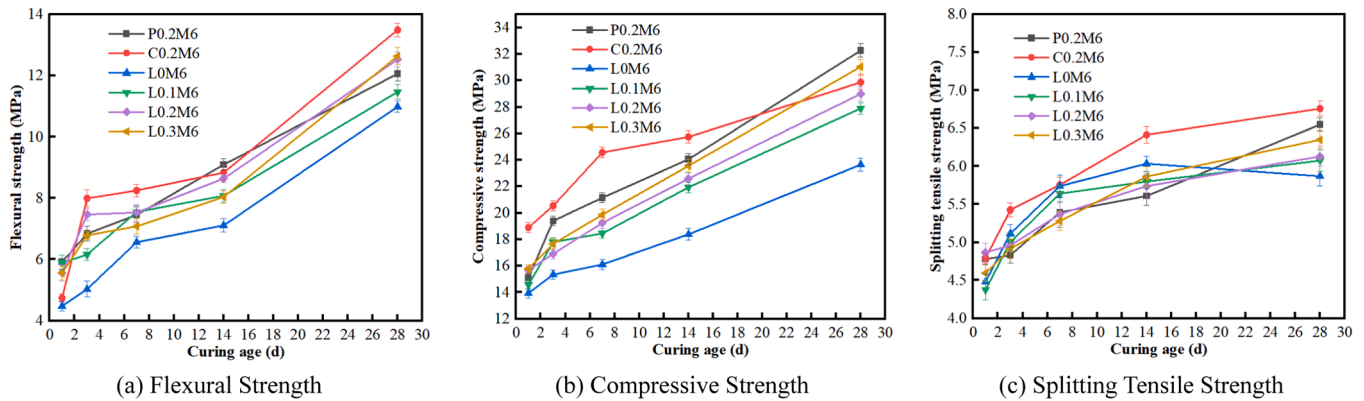


Fig. 14. Effect of Admixture Type and Dosage on the Mechanical Properties of BMSC-MS (M/S = 6).

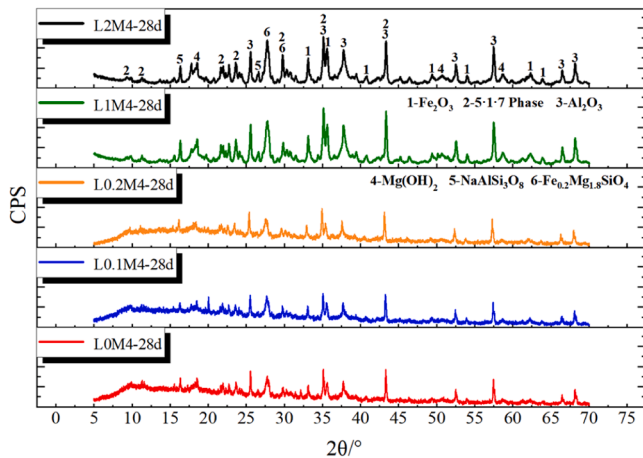


Fig. 15. XRD patterns of BMSC-MS with varying Lemon Powder dosages.

splitting tensile strengths. The P0.2M4 sample exhibited the highest 28-day compressive strength of 27.58 MPa, while L0.2M4 showed a compressive strength of 23.94 MPa, with all three strength indicators slightly lower than those of the other groups. These results indicate that citric acid, as the earliest systematically studied admixture in BMSC systems, provides a stable and consistent enhancement effect across different M/S ratios. Among the two target admixtures, phosphoric acid generally outperformed lemon powder in most performance indicators. Although the overall enhancement effect of lemon powder was relatively weak, it still demonstrated potential for performance improvement within a reasonable dosage range.

To further explore the adaptability of admixtures under different M/S ratios, Fig. 13 and Fig. 14 show the mechanical properties at M/S = 5 and M/S = 6, respectively. With increasing M/S ratio, the material strength gradually improved. For example, the 28-day compressive strength of P0.2M4 was 27.58 MPa; when the M/S ratio increased to 5 (P0.2M5), the compressive strength rose to 29.92 MPa, an increase of approximately 8.49%; at M/S = 6 (P0.2M6), the strength reached 32.28 MPa, an increase of about 17.06% compared with M/S = 4.

In the resource-constrained Martian environment, mix design optimization must balance mechanical performance with resource efficiency. Since MgO production requires high-temperature energy input, higher M/S ratios result in increased energy consumption. Therefore, although increasing the M/S ratio improves strength, the enhancement is relatively limited. From the perspective of engineering feasibility and resource conservation, M/S = 4 is considered the optimal choice that meets structural performance requirements.

Additionally, the simulated Martian soil contains various active components that can adsorb or chelate with admixtures, causing partial consumption of the active ingredients at low dosages and weakening their ability to regulate hydration reactions and microstructure evolution. Consequently, mechanical strength improvement is not obvious. To compensate for this effect, the dosages of phosphoric acid and lemon powder were increased to 1% and 2%, respectively, under M/S = 4 conditions.

As shown in Fig. 13, the compressive strengths of L1M4 and L2M4 increased to 34.90 MPa and 35.15 MPa, corresponding to enhancements of approximately 45.9% and 47% relative to L0.2M4. The compressive strengths of P1M4 and P2M4 were 39.09 MPa and 41.23 MPa, representing increases of 41.7% and 49.5% compared with P0.2M4. These results indicate that increasing the admixture dosage from 0.2% to 1% significantly improved compressive, flexural, and splitting tensile

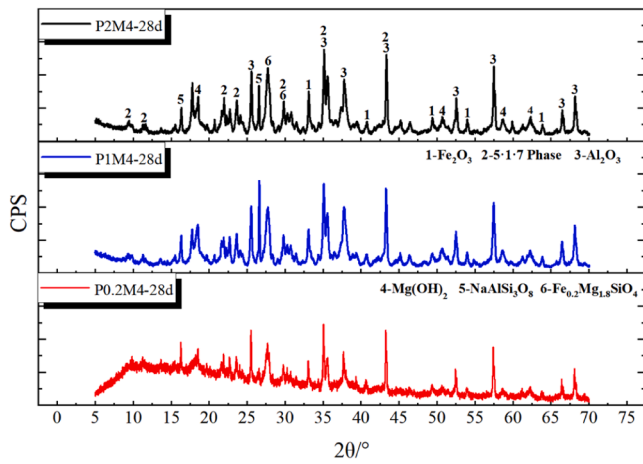


Fig. 16. XRD patterns of BMSC-MS with varying Phosphoric Acid dosages.

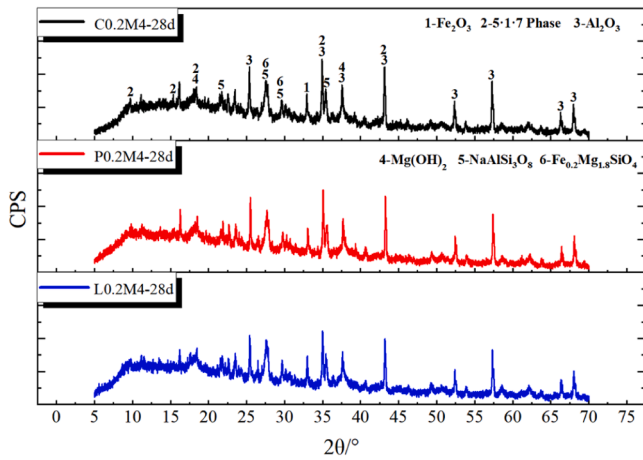


Fig. 17. Effects of different additive types on the XRD patterns of BMSC-MS.

strengths, with the most pronounced increase observed in compressive strength. Further increasing the dosage to 2% resulted in a plateau effect, suggesting that 1% is the optimal admixture dosage. Overall, BMSC-MS with 1% phosphoric acid exhibited the best mechanical performance and can be considered the optimal formulation.

4.3. Effects of different additive types and dosages on hydration products and microstructure of BMSC-MS

4.3.1. Phase composition analysis by XRD

Fig. 15 presents the X-ray diffraction (XRD) patterns of BMSC-MS systems with varying dosages of Lemon Powder. The major hydration products are identified as $\text{Mg}(\text{OH})_2$ and the 5-1-7 crystalline phase, with $\alpha\text{-Al}_2\text{O}_3$ introduced as an internal standard (20 wt%) for quantitative analysis of amorphous content. As the Lemon Powder dosage increases from 0.1% to 2%, the characteristic diffraction peaks of the 5-1-7 phase gradually intensify, and the crystalline structure becomes more well-defined, suggesting that the additive may facilitate crystallization and crystal growth.

At low additive dosages (L0.1M4 and L0.2M4), the 5-1-7 peaks are relatively weak, which may be attributed to complexation or adsorption interactions between active oxides in the Martian regolith simulant (Fe, Si, Al) and the carboxyl groups in Lemon Powder. These interactions may affect the availability of Mg^{2+} for the crystallization process, inhibiting nucleation and growth of the 5-1-7 phase, producing a loosely structured product and limiting macroscopic mechanical performance. When the dosage increases to 1% and 2%, the 5-1-7 phase is significantly enhanced, suggesting that sufficient additive can effectively regulate Mg^{2+} migration pathways, suppress disordered $\text{Mg}(\text{OH})_2$ precipitation, and promote the formation of ordered crystals, thereby improving microstructural density and mechanical properties [39].

Fig. 16 shows the XRD patterns of BMSC-MS with varying Phosphoric Acid dosages. With increasing Phosphoric Acid content, the diffraction peaks of the 5-1-7 phase continuously increase while the $\text{Mg}(\text{OH})_2$ peaks

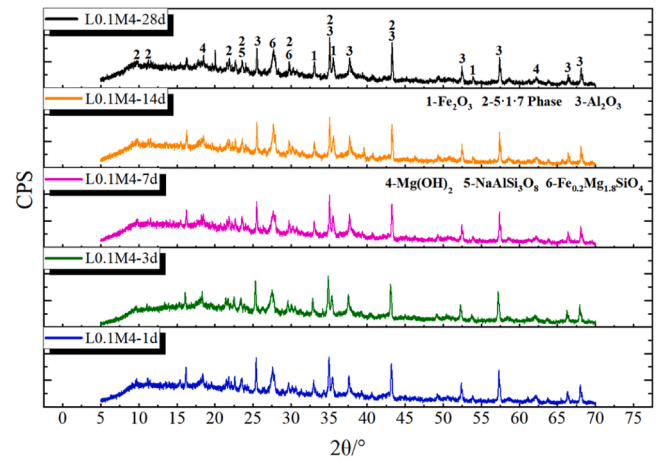


Fig. 18. XRD patterns of the L0.1M4 sample at different curing ages.

Table 7

Quantitative analysis of phase composition in BMSC-MS under different additive types and dosages.

No.	Age/d	Mass fraction/%									Rwp /%
		5-1-7 Phase	Mg(OH) ₂	MgCO ₃	MgO	Fe ₂ O ₃	Feldspar	Pyroxene	Olivine	Amor.	
L0.1M4	1d	3.8	14	0.2	2.4	2.9	30.76	6.43	2.01	17.5	11.61
L0.1M4	3d	5.3	13.9	0.5	1.8	3.1	30.65	6.21	1.94	16.6	10.47
L0.1M4	7d	6.3	13.1	0.7	1.6	3.5	28.12	5.70	1.78	19.2	11.28
L0.1M4	14d	6.6	12.9	1.0	0.8	3.0	26.73	5.01	1.57	22.4	13.62
L0.1M4	28d	7.5	12.2	1.4	0.5	2.3	26.78	3.82	1.51	24	12.43
L0.2M4	28d	8.8	11.2	1.6	0.3	3.9	25.10	3.88	1.53	23.7	11.16
C0.2M4	28d	10.0	10.3	1.2	0.8	3.5	25.07	2.67	1.46	25	10.29
P0.2M4	28d	11.5	9.2	1.5	0.4	3.4	25.41	2.94	1.55	24.1	10.61
P0.2M5	28d	11.9	8.9	1.4	0.5	3.0	25.02	2.86	1.52	24.9	10.23
P0.2M6	28d	12.6	8.3	1.6	0.0	2.9	25.62	2.78	1.50	24.7	9.95
P1M4	28d	14.5	6.9	1.5	0.6	3.6	25.59	1.58	1.43	24.3	14.29
P2M4	28d	18.3	4.0	1.7	0.5	3.0	25.99	1.66	1.46	23.4	10.23
L1M4	28d	13.1	7.9	1.6	0.8	4.1	25.94	2.85	1.52	22.2	14.83
L2M4	28d	14.4	7.0	2.0	0.4	4.3	25.46	2.75	1.49	22.2	13.81

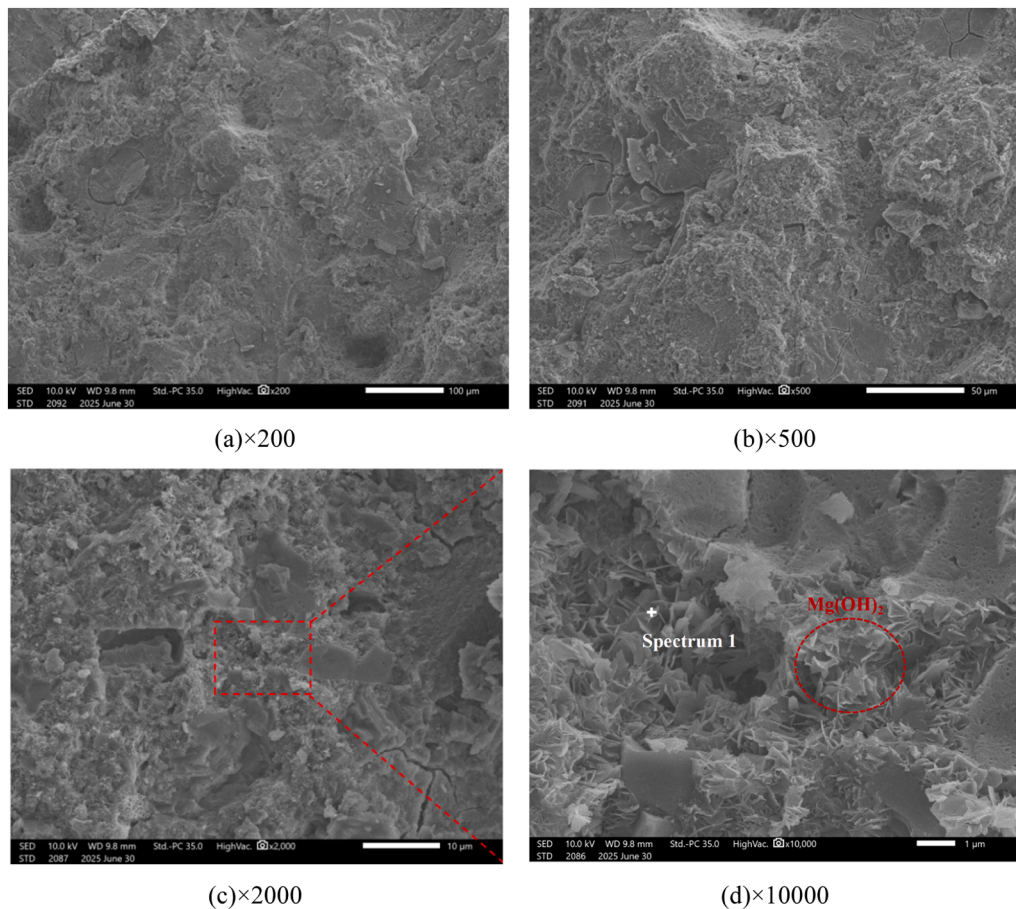


Fig. 19. Microstructural morphology of L0.2M4 sample at 28-day curing age.

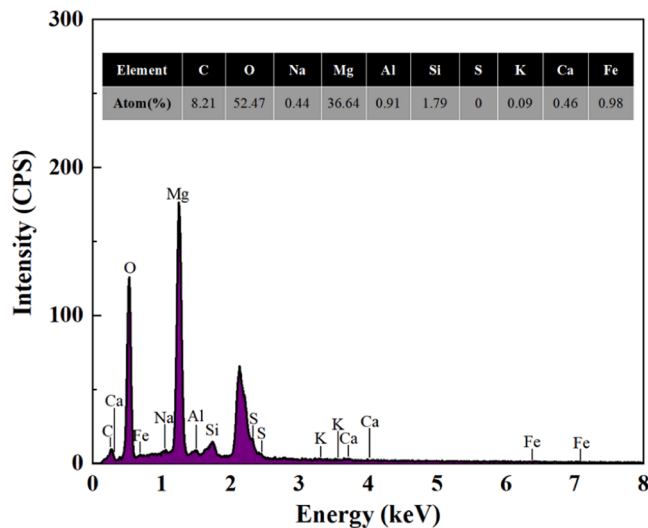


Fig. 20. EDS spectrum at Spectrum 1 of the BMSC-MS sample.

diminish, suggesting a more effective regulation of the reaction pathway by Phosphoric Acid. This trend may be associated with possible interactions between phosphate species and Mg^{2+} under alkaline conditions, which may help stabilize Mg^{2+} distribution and delay $Mg(OH)_2$ formation, thereby facilitating the crystallization of the 5-1-7 phase and improving structural density and stability [40].

Fig. 17 compares the XRD patterns of BMSC-MS with Phosphoric Acid, Citric Acid, and Lemon Powder at identical dosage and M/S ratio.

All three additives promote 5-1-7 formation, but the Phosphoric Acid system exhibits the most pronounced peaks and a more complete crystal structure. Differences in performance enhancement among the additives are attributed to their chemical structures and active components. Citric Acid, with its multi-carboxyl structure, exhibits strong chelating ability and can effectively regulate pore structure, improving microstructural integrity. Phosphoric Acid may contribute to a more favorable crystallization environment during the formation of Mg^{2+} concentration gradients, which may facilitate more ordered crystal growth and improve the formation and spatial distribution of the 5-1-7 phase [17,41]. In contrast, Lemon Powder is a complex natural extract mainly consisting of Citric Acid, lemon oil, and lemon juice, with relatively low content of active components and limited regulatory capability.

To quantify the phase assemblage in the BMSC-MS system, Rietveld refinement was performed using Topas software, and the calculated results are summarized in Table 7. Fig. 18 presents the XRD patterns of the L0.1M4 specimen at different curing ages. As shown in the figure and table, with increasing hydration age, the content of the 5-1-7 phase increased steadily from 3.8% at 1 d to 7.5% at 28 d, indicating its continuous formation and development throughout the hydration process. In contrast, the $Mg(OH)_2$ content gradually decreased from 14.0% at 1 d to 12.2% at 28 d. This trend is associated with the complexation regulation of Mg^{2+} by the additive: the carboxyl groups in citric powder partially chelate Mg^{2+} , suppressing its disordered precipitation as $Mg(OH)_2$ and thereby providing more available Mg^{2+} for the ordered nucleation and growth of the 5-1-7 phase.

The amorphous content exhibited an initial decrease followed by a subsequent increase. From 1 d to 3 d, it decreased from 17.5% to 16.6%, corresponding to the transformation of early-stage amorphous precursors (low-order 5-1-7 phase and primary $Mg(OH)_2$ gel) into

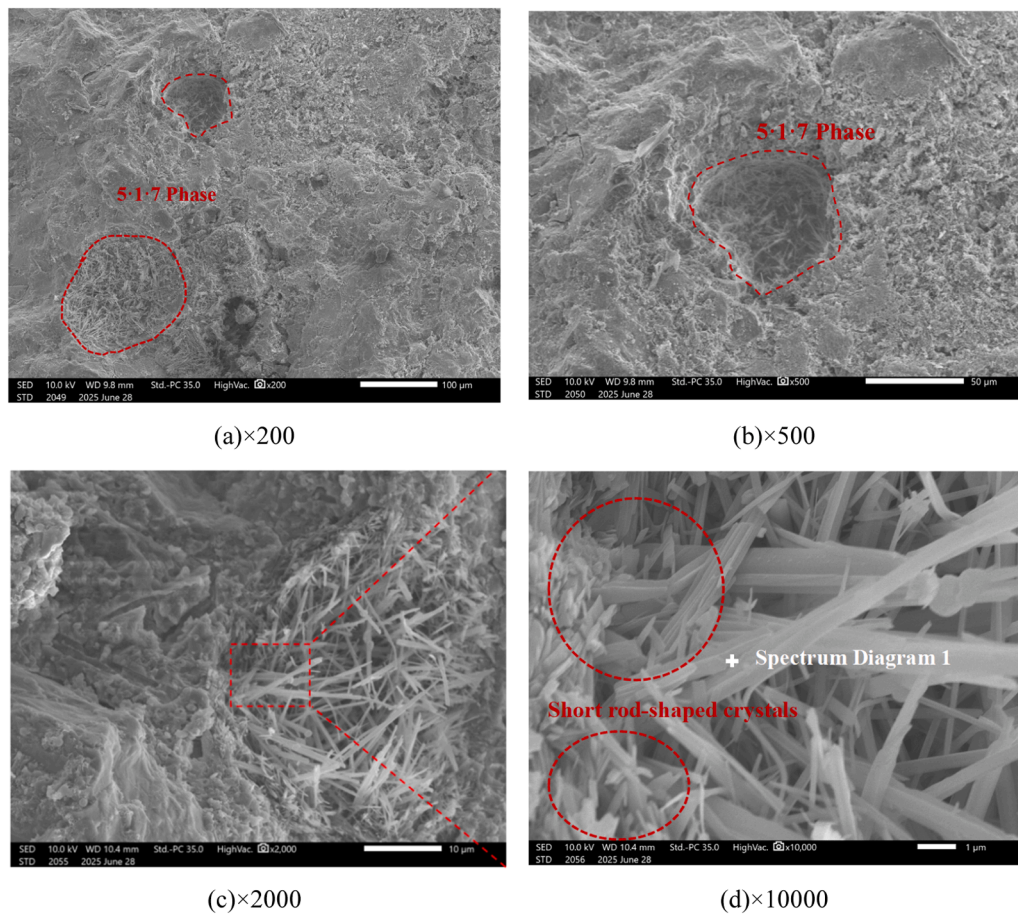


Fig. 21. Microstructural morphology of P0.2M4 sample at 28-day curing age.

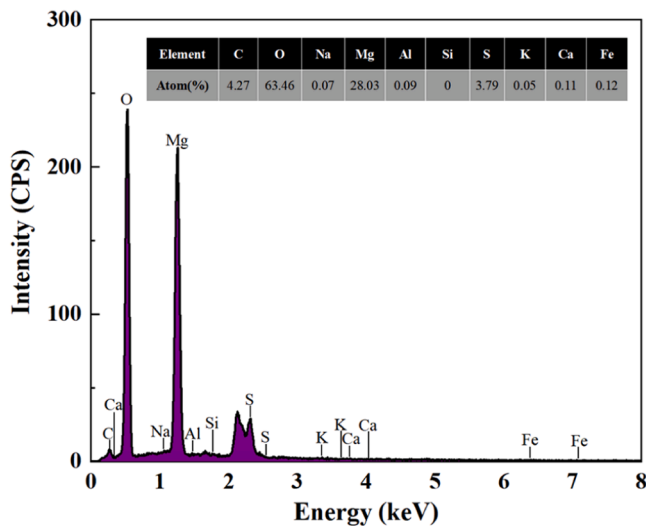


Fig. 22. EDS spectrum at Spectrum 2 of the BMSC-MS sample.

crystalline phases. From 3 d to 28 d, the amorphous fraction increased from 16.6% to 24.0%, which can be attributed to the sustained pozzolanic reaction of reactive aluminosilicate components in the regolith under alkaline conditions, leading to the formation of predominantly amorphous gels such as M-S-H and M-A-S-H [27]. Although these amorphous gels do not exhibit distinct diffraction peaks, they effectively fill pores and enhance structural densification, thereby contributing substantially to strength development [42].

Moreover, the diffraction baselines in the low-angle region (5° – 10°) of the samples in Figs. 15 and 16 display notably distinct morphological features. Such variations stem primarily from the microstructural evolution of the matrix under different admixture dosages. Specifically, higher admixture dosages not only further facilitate the crystallization and grain refinement of the 5-1-7 phase [39,40], but also alter the spatial distribution and local structure of the amorphous gels derived from the reactive Si/Al components in the Martian regolith [27]. These microstructural modifications thereby alter the diffuse scattering behavior at low angles, ultimately resulting in the observed morphological differences in the baselines [43].

Note: Rwp, the R-weighted pattern, represents the quantitative fitting error calculated by Topas software.

4.3.2. Microstructure analysis by SEM-EDS

To further elucidate the regulatory mechanism of additive type and dosage on the microstructure of the BMSC-MS system, four 28-day specimens, namely L0.2M4, P0.2M4, P1M4, and P2M4, were selected for SEM observation, which was further verified by EDS analysis. During the test, five representative EDS spectra were specifically selected, corresponding to five key characteristic regions during the microstructural evolution of the system, namely the initially formed plate-like $\text{Mg}(\text{OH})_2$ (Spectrum 1), the initially formed short rod-like 5-1-7 crystals (Spectrum 2), the well-developed columnar 5-1-7 crystal clusters (Spectrum 3), the pore-filling amorphous M-(A)-S-H gel (Spectrum 4), and the 5-1-7 crystals deformed by crystallization pressure (Spectrum 5).

Fig. 19 presents the microstructural morphology of L0.2M4. As shown in Figs. 19(a) and 19(b), no obvious 5-1-7 crystals were observed at low magnification. When further magnified, a large amount of plate-like hydration products can be clearly observed on the surface, as

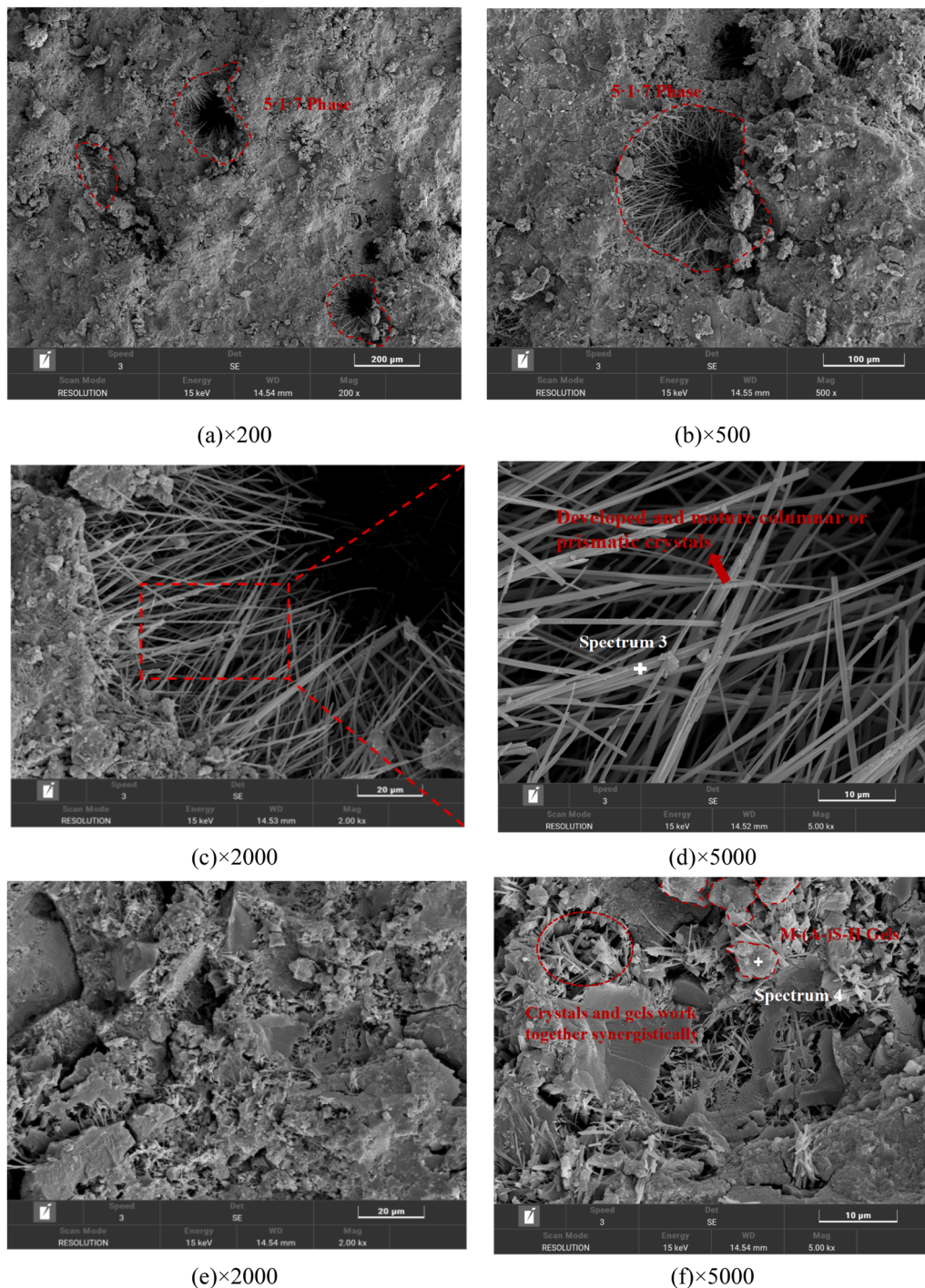


Fig. 23. Microstructural morphology of P1M4 sample at 28-day curing age.

marked by the red circle in Fig. 19(d). The EDS analysis shown in Fig. 20 (Spectrum 1) indicates that the main elements are Mg and O, which can be identified as $\text{Mg}(\text{OH})_2$. The overall structure is relatively loose and lacks elongated columnar or prismatic crystals. In the system containing only citric powder, no significant growth of 5-1-7 crystals was observed, whereas a large amount of flaky $\text{Mg}(\text{OH})_2$ products formed. This may be attributed to the relatively weak complexation between the carboxyl groups in citric powder and Mg^{2+} under the present alkaline conditions, thereby limiting its ability to inhibit the rapid precipitation of $\text{Mg}(\text{OH})_2$. Meanwhile, under the regulation of lemon powder, the reaction of active Si/Al components in the Martian regolith was not sufficiently activated, leading to insufficient gel formation and relatively high microstructural

porosity. This result is consistent with the weak 5-1-7 diffraction peaks and relatively high amorphous content observed in the XRD results, indicating that the strengthening mechanism is mainly dominated by magnesium hydroxide, while the contribution from 5-1-7 crystals is relatively limited.

Fig. 21 shows the microstructural morphology of P0.2M4. At low magnification, as shown in Figs. 21(a) and 21(b), 5-1-7 crystals can already be observed in some local regions. At higher magnification, several short rod-like crystals are found growing inside the pores, as marked by the red circle in Fig. 21(d). The EDS result in Fig. 22 (Spectrum 2) further confirms that these crystals belong to the 5-1-7 phase. This indicates that at a phosphoric acid dosage of 0.2%, phosphoric acid

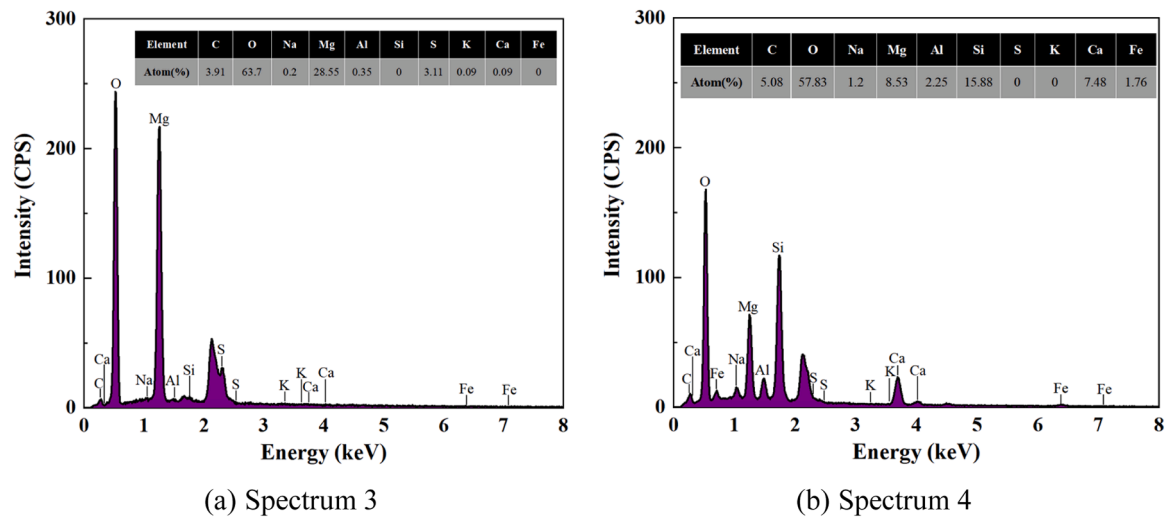


Fig. 24. EDS spectrum at Spectrum 3 and Spectrum 4 of the BMSC-MS sample.

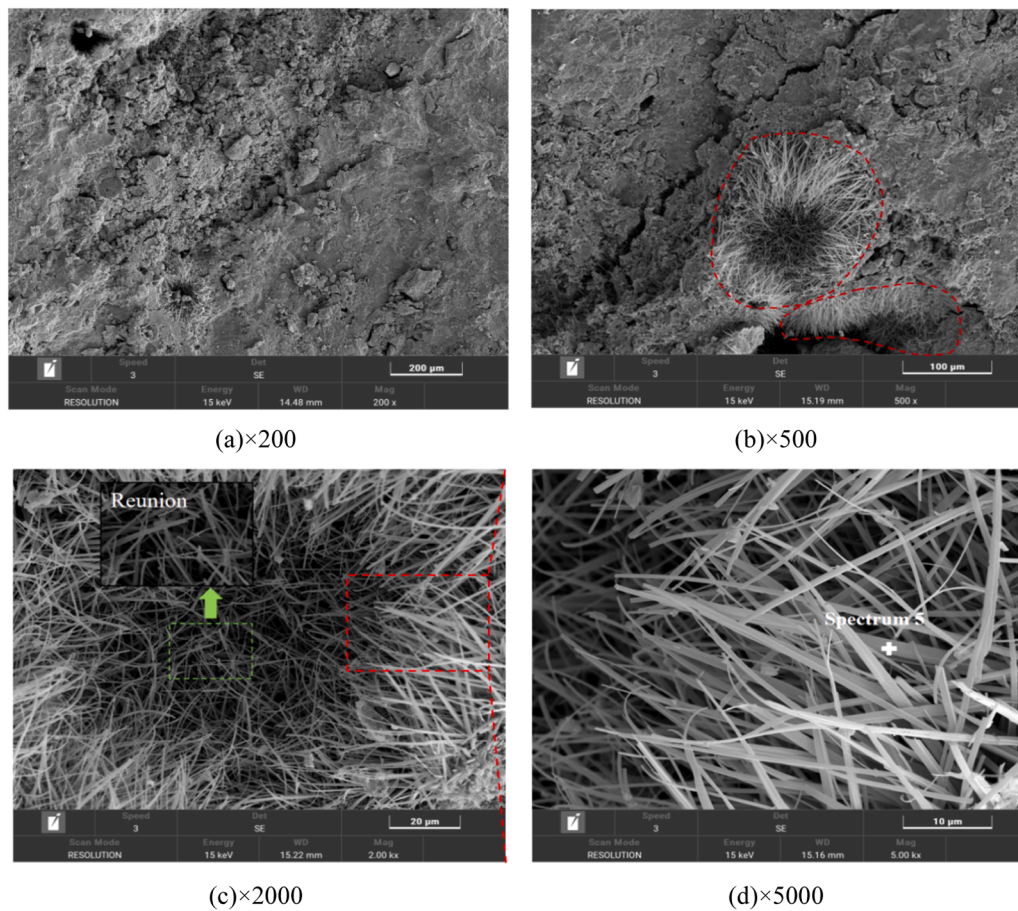


Fig. 25. Microstructural morphology of P2M4 sample at 28-day curing age.

has already begun to effectively complex part of the Mg^{2+} ions and induce the nucleation and growth of 5-1-7 crystals. Combined with the XRD results, the intensity of the 5-1-7 diffraction peak is significantly enhanced, and the mechanical performance is also markedly higher than that of L0.2M4, further confirming the role of phosphoric acid in promoting nucleation and synergistic crystal growth.

Fig. 23 presents the microstructural morphology of P1M4. At low magnification, clustered 5-1-7 crystals can already be observed in the

pores, as shown in Figs. 23(a) and 23(b). At higher magnification, a large number of well-developed columnar or prismatic crystals can be clearly observed, as indicated by the red arrows in Fig. 23(d). In some regions, densely arranged crystal cluster structures are formed, with regular crystal faces and ordered orientation, indicating that the increased phosphoric acid dosage promotes the growth of 5-1-7 crystals. The EDS result in Fig. 24(a) (Spectrum 3) further confirms that these crystals belong to the 5-1-7 phase, further demonstrating the promoting effect of

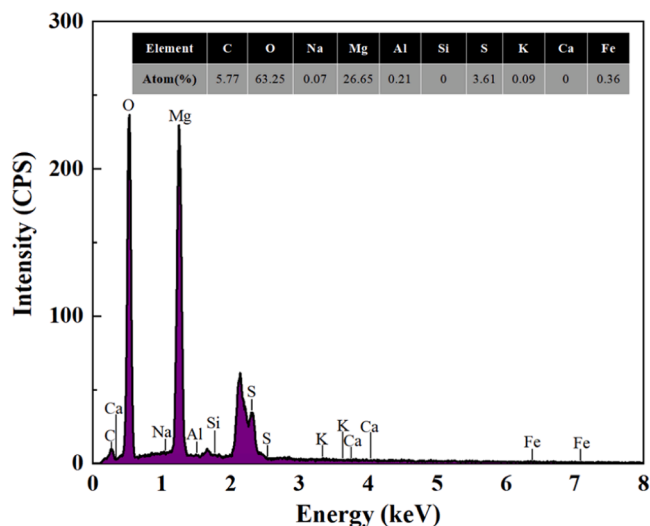


Fig. 26. EDS spectrum at Spectrum 5 of the BMSC-MS sample.

phosphoric acid on crystal growth. Meanwhile, obvious flocculent gel products can be observed in the interwoven gaps between crystals, as marked by the red circle in Fig. 23(f). The EDS analysis in Fig. 24(b) (Spectrum 4) shows that the main elements are Mg, Si, and O, which can be inferred as M-(A)-S-H gel [27]. This observation shows that, in the presence of phosphoric acid, well-developed 5-1-7 crystals and flocculent M-(A)-S-H gel coexist. As previously reported [27], the Martian regolith simulant can generate M-(A)-S-H gel under alkaline BMSC conditions. Here, the gel is observed to fill the intercrystalline spaces of the 5-1-7 phase, which is consistent with a densified microstructure. In addition, the glassy phase in the volcanic ash is activated under alkaline conditions to generate M-(A)-S-H gel, which also provides a more suitable pH environment for the stable formation of the 5-1-7 phase. Therefore, it can be seen that the P1M4 specimen forms a composite microstructure with synergistic crystal-gel interaction under phosphoric acid induction, which not only ensures skeleton support but also realizes pore filling, thereby significantly improving structural compactness and overall stability. This feature is highly consistent with the enhanced 5-1-7 diffraction peak and improved mechanical properties observed in the XRD results, further indicating that an appropriate amount of phosphoric acid can achieve the optimal balance between crystal growth and gel formation.

Fig. 25 presents the microstructural characteristics of P2M4. At low magnification, clustered 5-1-7 crystals can still be observed in the pores, as shown in Figs. 25(a) and 25(b). At higher magnification, a large number of well-developed columnar or prismatic crystals can be observed, accompanied by interlaced crystal growth, as shown in the green box in Fig. 25(c). Previous studies have shown that additives can promote the nucleation and growth of the 5-1-7 phase in the BMSC system by changing the ionic concentration field and reaction kinetics. However, when the additive dosage or local ionic concentration exceeds a certain threshold, the reaction process may shift from “gradual densification” to a “rapid crystallization-dominated” mode. Under such conditions, crystals rapidly grow within the relatively dense matrix pores and are mutually compressed due to spatial constraints, resulting in internal crystallization stress and inducing microcracking or pore redistribution [44]. Meanwhile, the rapid crystallization process is prone to generating irregular intergrown or bridged crystals, which disrupt the original interlocking crystal structure and further concentrate local stress, manifested as microcracks or local spalling. Based on the above observations and analysis, the coexistence of extensive crystal nucleation/growth and local structural defects in P2M4 can be reasonably explained. Therefore, the additive dosage should be controlled within a range that can promote the formation of target products

without inducing excessive crystallization stress, so as to achieve the coordinated optimization of microstructural densification and macroscopic mechanical performance.

4.3.3. TG-DSC analysis

From the above analyses, it is evident that excessive addition of phosphoric acid not only leads to resource waste under Martian conditions but also provides limited enhancement to the performance of the BMSC-MS system. Therefore, subsequent analyses do not include the P2M4 sample.

Fig. 27 presents the DSC-TG curves of L0.2M4, P0.2M4, and P1M4 samples at 28 days. In the DSC curves, the first endothermic valley appears near 95 °C for all three samples, accompanied by significant mass loss on the TG curves. This stage corresponds to the removal of adsorbed and interlayer water from M-(A)-S-H gels [45], while the 5-1-7 phase ($5\text{Mg}(\text{OH})_2 \cdot \text{MgSO}_4 \cdot 7\text{H}_2\text{O}$) begins to lose two molecules of crystallization water, transforming into the 5-1-5 phase (as shown in Eq. (4)) [19].

Upon further heating, a second prominent endothermic valley appears around 141 °C, corresponding to continued mass loss on the TG curve. This is mainly attributed to the 5-1-5 phase losing an additional two molecules of crystallization water to form the 5-1-3 phase (Eq. (5)) [19]. Notably, the peak of P1M4 is the most pronounced, indicating a higher proportion of 5-1-7 crystals and a more complete hydration process. Between 141 and 250 °C, the residual crystallization water in the 5-1-3 phase is completely removed, yielding an anhydrous basic magnesium sulfate phase, $5\text{Mg}(\text{OH})_2 \cdot \text{MgSO}_4$ (Eq. (6)) [19]. Additionally, a secondary endothermic peak appears near 250 °C for all three samples, accompanied by slight mass loss, corresponding to the release of structural water from M-(A)-S-H gels [45,46], during which the amorphous phases undergo further depolymerization and shrinkage.

It is noteworthy that L0.2M4 exhibits a distinct exothermic peak near 200 °C, with minor mass loss on the TG curve, attributed to combustion of organic components in the lemon powder additive. In contrast, P0.2M4 and P1M4 do not show such features, indicating that the reaction mechanism in the phosphoric acid system is dominated by inorganic phase rearrangement. From the TG curves, the mass loss during the first stage (room temperature to 300 °C) is concentrated and significant, with P1M4, P0.2M4, and L0.2M4 showing 9.92%, 9.86%, and 8.76%, respectively, reflecting the differing contributions of the additives and their dosages to hydration and product formation.

Around 376 °C and 455 °C, two strong endothermic valleys appear in the DSC curves, corresponding to significant mass loss on the TG curves, primarily due to the decomposition of $\text{Mg}(\text{OH})_2$ (Eq. (7)), marking the disruption of $\text{Mg}(\text{OH})_2$ stability and dominating the second stage of TG mass loss. Upon further heating to 500–600 °C, broad endothermic valleys are observed, corresponding to the removal of deep structural water from M-(A)-S-H gels. During this process, the gel framework gradually collapses, forming MgO and amorphous $\text{SiO}_2/\text{Al}_2\text{O}_3$ (Eq. (8)) [47], indicating complete transformation of the amorphous phases at high temperature.

Near 600 °C, all three samples show minor endothermic valleys with slight mass loss, corresponding to MgCO_3 decomposition (Eq. (9)). The weak intensity of these peaks suggests limited carbonate formation, consistent with XRD quantitative analysis. Moreover, in the 300–400 °C range, P0.2M4 and P1M4 display additional exothermic peaks, likely due to rearrangement of Mg^{2+} and PO_4^{3-} during heating, forming stable solid magnesium phosphate or complex salt structures.

When the temperature reaches 700–1000 °C, the TG curves enter the third stage, exhibiting pronounced mass loss accompanied by sharp endothermic valleys in the DSC curves. This stage corresponds to MgSO_4 decomposition with SO_3 release (Eq. (10)) [19], indicating that the system has reached the final high-temperature phase reconstruction stage.



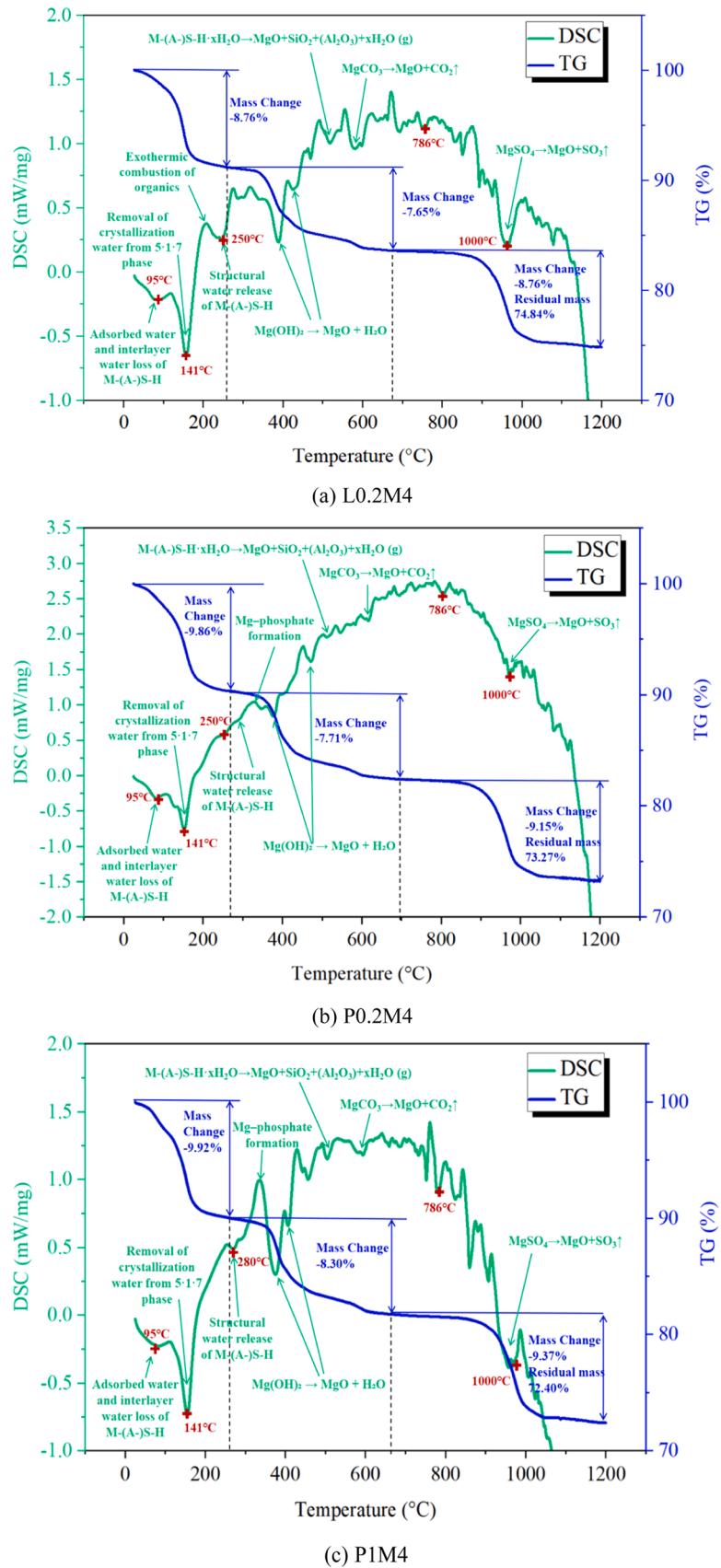


Fig. 27. DSC-TG curves of L0.2M4, P0.2M4, and P1M4 samples at 28 days. Note: The red crosses in the figure indicate the dehydration temperature points of the 5-1-7 phase at different stages, providing a visual representation of its decomposition characteristics.

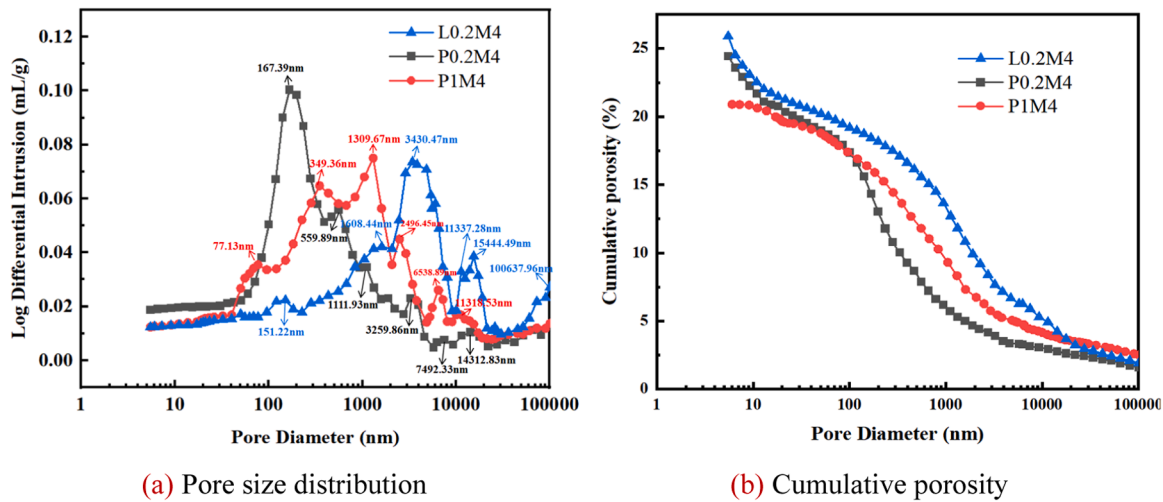
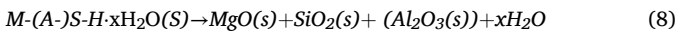
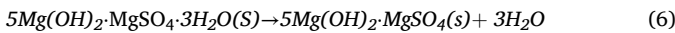
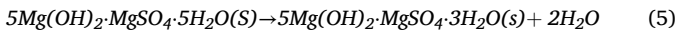


Fig. 28. Pore size distribution and cumulative porosity of L0.2M4, P0.2M4, and P1M4 samples at 28 d.

Table 8

Pore structure test results of different BMSC-MS samples.

Sample	Average pore diameter/nm	Porosity/%	Pore size distribution/%			
			< 10 nm	10 nm~100 nm	100 nm~1000 nm	> 1000 nm
L0.2M4	41.09	25.90	13.02	12.90	21.28	52.80
P0.2M4	42.27	24.43	11.26	17.72	45.72	25.30
P1M4	123.94	20.89	1.34	15.36	38.80	44.50



4.3.4. Pore structure analysis by MIP

According to pore size, the pores can be classified as gel pores (<10 nm), transition pores (10–100 nm), capillary pores (100–1000 nm), and macropores (>1000 nm) [48]. Fig. 28 shows the pore size distribution and porosity of L0.2M4, P0.2M4, and P1M4 specimens, and Table 8 summarizes the pore volume fractions in different size ranges.

The most probable pore diameter refers to the pore size corresponding to the peak in the differential curve, i.e., the diameter with the highest occurrence probability [49]. As shown in Fig. 28(a) and Table 8, L0.2M4 exhibits six most probable pores (151.22 nm, 1608.44 nm, 3430.47 nm, 11337.28 nm, 15444.49 nm, and 100637.96 nm), mainly within the capillary and macropore ranges, with macropores accounting for 52.80%, resulting in a loose microstructure. P0.2M4 shows six most probable pores (167.39 nm, 559.89 nm, 1111.93 nm, 3259.86 nm, 7492.33 nm, and 14312.83 nm), also distributed in the capillary and macropore regions, but with macropore fraction reduced to 25.30%. This indicates that low-dose phosphoric acid promotes crystal formation and partially fills large pores, lowering total porosity from 25.90% to 24.43%. In comparison, P1M4 exhibits six most probable pores spanning transition, capillary, and macropore regions (77.13 nm, 349.36 nm, 1309.67 nm, 2496.45 nm, 6538.89 nm, and 11318.53 nm), with a wider and more balanced pore size distribution, total porosity further reduced to 20.89%, while the average pore diameter increases to 123.94 nm.

As shown in Table 8, the average pore diameter follows the trend P1M4 > P0.2M4 > L0.2M4, whereas total porosity follows L0.2M4 > P0.2M4 > P1M4. This can be explained by crystal growth, pore evolution, and gel-filling mechanisms. L0.2M4 has limited crystal formation and insufficient gel filling, with more than half of the pores being macropores, resulting in a loose structure and incomplete hydration product filling. P0.2M4, under low-dose phosphoric acid regulation, exhibits increased crystal content and partial filling of macropores, reducing total porosity. However, the microstructure is still gel-dominated, yielding a relatively small average pore diameter of 42.27 nm. The distribution of the six most probable pores shows that P0.2M4's peaks remain in the capillary and macropore ranges and are generally larger than the corresponding peaks in P1M4, indicating that low-dose phosphoric acid, while promoting crystal precipitation and reducing some macropores, does not significantly refine pore size or achieve uniform distribution, limiting the enhancement of overall densification.

In P1M4, phosphoric acid significantly promotes 5-1.7 crystal formation, with crystals clustered or orderly arranged within the pores. The interstitial spaces form numerous transition (10–100 nm) and capillary (100–1000 nm) pores, increasing the average pore diameter. Simultaneously, newly formed crystals fill original macropores, lowering total porosity and densifying the microstructure. Transition pores not only facilitate moisture diffusion for continued hydration but also provide space for crystal-gel synergistic densification, maintaining structural integrity while reducing stress concentration. This leads to a more uniform pore size distribution, decreased macropore proportion, and introduction of additional small pore peaks, resulting in the lowest total porosity and the best densification performance.

5. Conclusions

(1) A conceptual ISRU-oriented preparation pathway for Martian BMSC was proposed, and its material-level feasibility was preliminarily evaluated under terrestrial laboratory conditions, covering the

extraction of reactive MgO from hydrated magnesium sulfate minerals, the production of phosphoric acid from phosphate minerals, and the final preparation of cement and concrete. The proposed route demonstrates high potential for in-situ resource utilization.

(2) Both phosphoric acid and lemon powder improved the mechanical properties of the BMSC-MS system, while phosphoric acid exhibited a more pronounced enhancement effect. The optimum performance was achieved at an M/S molar ratio of 4 with 1 wt% phosphoric acid, reaching a 28-day compressive strength of 39.09 MPa.

(3) Phosphoric acid significantly promoted the formation and crystallization of the 5-1.7 phase and effectively suppressed the disordered precipitation of $\text{Mg}(\text{OH})_2$. Meanwhile, it activated the reactive Si/Al components in the simulated Martian regolith, leading to the formation of amorphous M-(A)-S-H gels and a denser composite microstructure.

(4) The incorporation of 1 wt% phosphoric acid effectively refined the pore structure and reduced the total porosity to 20.89%, while also improving the thermal stability of the BMSC-MS system.

(5) The simulated Martian regolith exhibited evident pozzolanic reactivity in the alkaline BMSC environment, continuously generating M-(A)-S-H gels that contributed to later-age strength development and microstructural densification.

CRedit authorship contribution statement

Lu Mingyang: Investigation. **Kang Yanqi:** Investigation. **Liu Weifeng:** Investigation. **Zhao Zonghao:** Investigation. **Li Lingyu:** Writing – original draft, Visualization, Validation, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Ma Haiyan:** Supervision, Resources, Project administration. **Yu Hongfa:** Writing – review & editing, Supervision, Project administration, Methodology, Conceptualization. **Zhang Honglei:** Visualization, Investigation. **Zhao Yanbing:** Writing – review & editing, Supervision, Project administration, Methodology, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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